

Tylus Dawkins

Full-Stack Engineer

725-500-3010 | TylusJDawkins@gmail.com | [linkedin.com/in/tylusedawkins](https://www.linkedin.com/in/tylusedawkins) | github.com/TylusDawkins

Full-Stack Engineer with direct production experience in P&C insurance at Liberty Mutual, building claims and billing automation in a regulated, compliance-first environment. Specializes in real-time systems, microservice architecture, and cloud infrastructure across AWS and Azure with a systems-level emphasis on auditability, correctness, and data migration at scale. Comfortable across the full stack and multiple languages; experienced shipping production software under regulatory constraints where correctness and long-term maintainability are non-negotiable.

Areas of Expertise

Languages & Frameworks: C#, .NET (Orleans, Blazor/Maui), JavaScript/TypeScript (React 18, Next.js 14, Solid.js, Node/Express), Python (Django), Tailwind CSS

Cloud & Data: AWS, Azure, Docker, Kubernetes, CI/CD, MSSQL, PostgreSQL, MySQL, MongoDB

Experience

Full Stack Software Engineer - 7/01/2025-Present - Circa Sports - Full time - US

- Developed a backend .NET Orleans and MSSQL sports betting application for managing real time data systems in a regulated environment.
- Used .NET Maui/Blazor to create a mobile/desktop application for sports betting, as well as designing and implementing the UI for content and customer service management system.
- Used Visual Basic and C# to create a real time WinForm UI for managing betting related objects such as events, sports, subsports, managing tickets withdrawals and deposits.
- Designed and implemented a user migration pipeline to safely transition 15,000+ accounts from a legacy system, handling partial failures, data validation, and recovery tooling to ensure integrity and continuity.
- Migrated features from legacy windows native and winform applications to .NET framework and Maui/Blazor applications.

Software Engineer - 06/01/2022-12/01/2023 - Liberty Mutual - Full Time - US

- Designed and implemented production features in a regulated insurance environment, emphasizing data correctness, auditability, automated testing, and unit tests.
- Developed multiple chat bots for claims, billing and more using Typescript/Node, AWS Lambdas, AWS EC2, AWS Lex and AWS Connect, increasing caller retention by 40%, and customer phone satisfaction by 15%.
- Pioneered automated Cyara IVR testing to proactively detect failures caused by upstream data and process changes, reducing production risk and improving system reliability.
- Created a Python/Django application for creating migratable firewall templates that worked for Azure ARM, Bicep, AWS CFT and terraform.
- Mentored 15+ new engineers in group and individual settings.

Instructor Associate - 10/01/2022-05/01/2025 - General Assembly - Part Time/Full time - US/KSA

- Taught full-stack development with JavaScript, TypeScript, React, Next.js, Node/Express, Python/Django, PostgreSQL, and MongoDB covering component architecture, state management, SSR vs. CSR rendering patterns, and RESTful API design.
- Emphasized testing methodology, cloud principles, the uses of containerization and clear system design to prepare students for production and regulated environments.
- Educated students on agile methodology consisting of daily standups, backlog refinement, sprint planning and refinement preparing them for collaborative work.
- Assisted in development of, graded and provided feedback for thousands of student projects.

- Maintained top 5% KPIs for instructional performance throughout my tenure at General Assembly.

Software Engineer - 04/01/2022-03/01/2024 - Doka - Contract/PT - US

- Built the UI for an internal delivery tracking and management system using TypeScript and Angular.
- Enabled migration from spreadsheet-based workflows to a live web application, improving usability and reliability.

Freelancing and Personal Projects

Two-Way Translator Real-Time AI Speech Translation Platform (Personal Project)

- Architected a multi-service real-time speech transcription and translation platform built for production-grade reliability and extensibility by designing with horizontal and vertical scaling in mind.
- SolidJS frontend with live WebSocket streaming for real-time transcript delivery; FastAPI backend handling JWT auth, and session management.
- Built 4 specialized microservices: authenticated audio ingest receiver, GPU-accelerated Whisper AI transcriber (large-v3), transcription merger managing text assembly and persistence across active sessions, and main session API.
- Redis pub/sub as async message broker between services; PostgreSQL with Alembic migrations for persistence; multi-profile Docker Compose deployment supporting both CPU and GPU hardware targets.

Design Your Jewelry Full-Stack Next.js E-Commerce Platform (Freelance)

- Built a custom jewelry design and e-commerce platform using Next.js 14 (Pages Router), Tailwind CSS, and PostgreSQL deployed to Vercel.
- Engineered a drag-and-drop design board using @dnd-kit supporting 7 chain configurations, per-anchor bead placement, real-time skin-tone mannequin preview, and local storage design persistence across sessions.
- Implemented dynamic routing, Next.js API Routes for RESTful endpoints, and server-side auth validation using Clerk's `getAuth()` returning 401/403 to prevent user spoofing at the API layer.
- Designed a Stripe payment integration and admin order management pipeline covering design submission, handcraft fulfillment tracking, and shipment management with a full Sequelize ORM data model and migrations.
- Applied SSR/CSR trade-offs deliberately: disabled SSR on the interactive design canvas for performance while keeping API routes server-rendered; integrated Vercel Blob for image storage and SWR for client-side data fetching.

Education

General Assembly 02/01/2022-05/01/2022 Software Engineering Immersive (420 course hours)